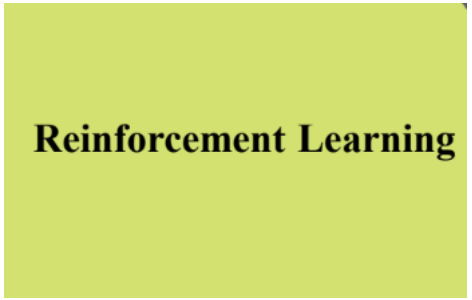
Computer Vision

DPS920

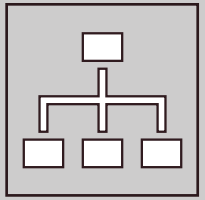
Session 14

ML Algorithm Categorizations



ML Algorithm Categorizations

Supervised Learning Algorithms Categorization



Classification

Regression

Regression

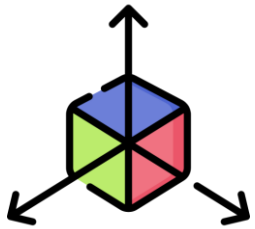
i) Simple

linear

nonlinear

i) Multiple

linear

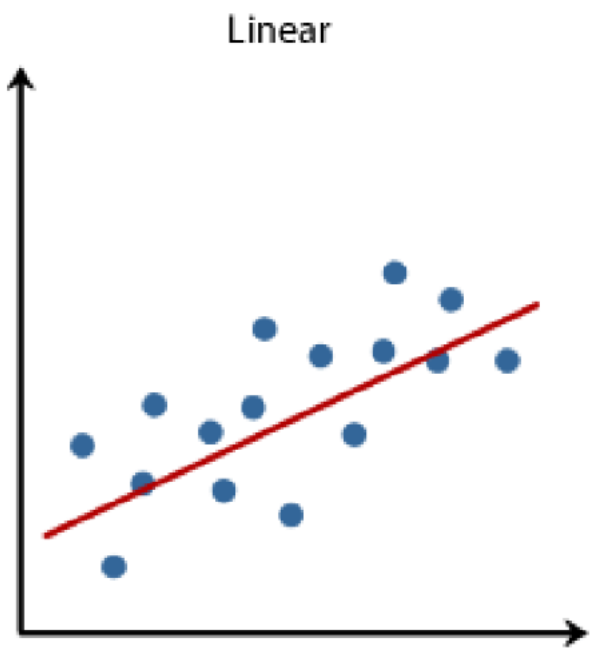


Polynomial

nonlinear

Simple Linear Regression

model the relationship between one independent variable (X) and one dependent variable (Y)

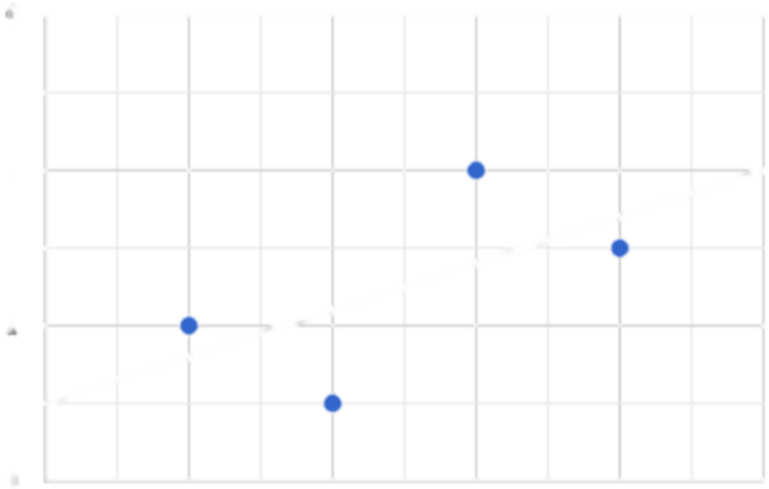


intercept

$$y = a + bx$$

slope

Problem



Our goal:

find a line that minimizes our error

Solution 1

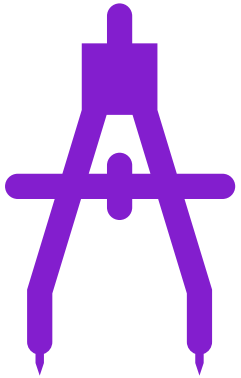
Visualize data and plot points → not scalable

Use Closed-Form Formulas → complex

Use optimization algorithms

Formulate

Points: (1,2), (2,1), (3,4), (4,3)



Solution: map to a linear line

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

Can't find the exact **linear** line

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array}$$

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array} \longrightarrow \text{Minimize the distance}$$

Solution

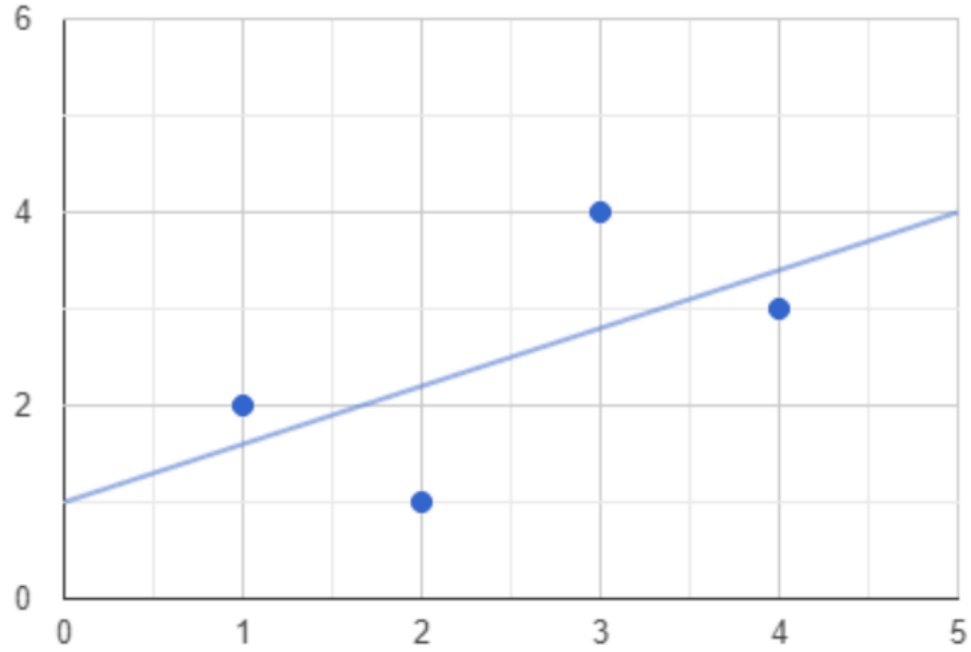
- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array} \longrightarrow \min \left[(y1-y1')^2 + (y2-y2')^2 + (y3-y3')^2 + (y4-y4')^2 \right]$$

Example



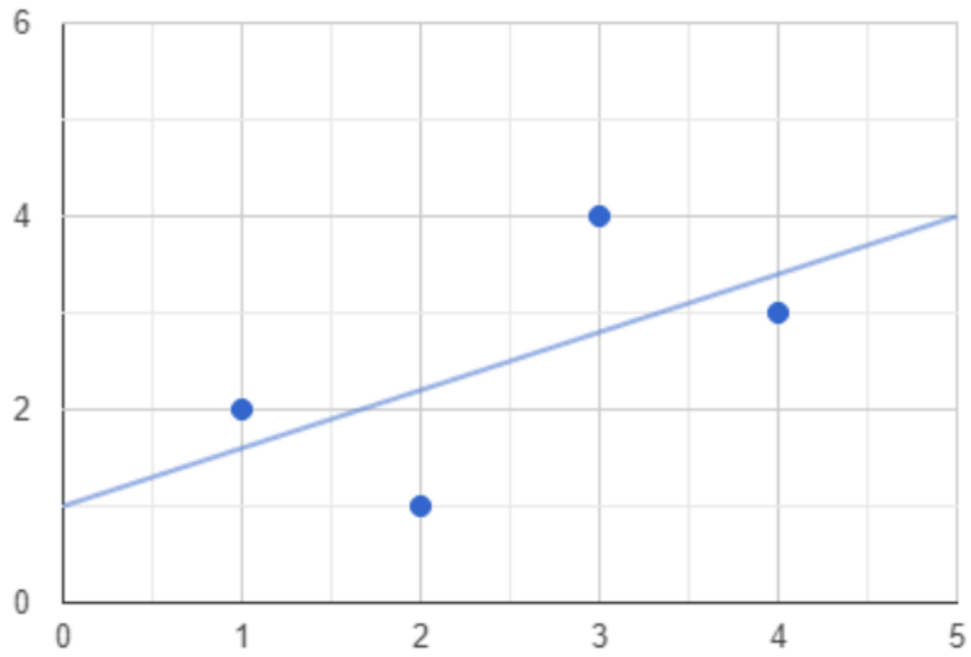
Actual outputs: 2, 1, 4, 3

Line's outputs: -1, 0, 1, 2

We do not know the equation of the line! But we want to find it

$y' = ax + b$

Example



Actual outputs: 2, 1, 4, 3

$$(y1-y1')^2 + (y2-y2')^2 + (y3-y3')^2 + (y4-y4')^2$$

$$\text{loss/distance} = (2-(a+b))^2 + (1-(2a+b))^2 + (4-(3a+b))^2 + (3-(4a+b))^2$$

Example

$$\text{loss/distance} = (2-(a+b))^2 + (1-(2a+b))^2 + (4-(3a+b))^2 + (3-(4a+b))^2$$

$$= 30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$$

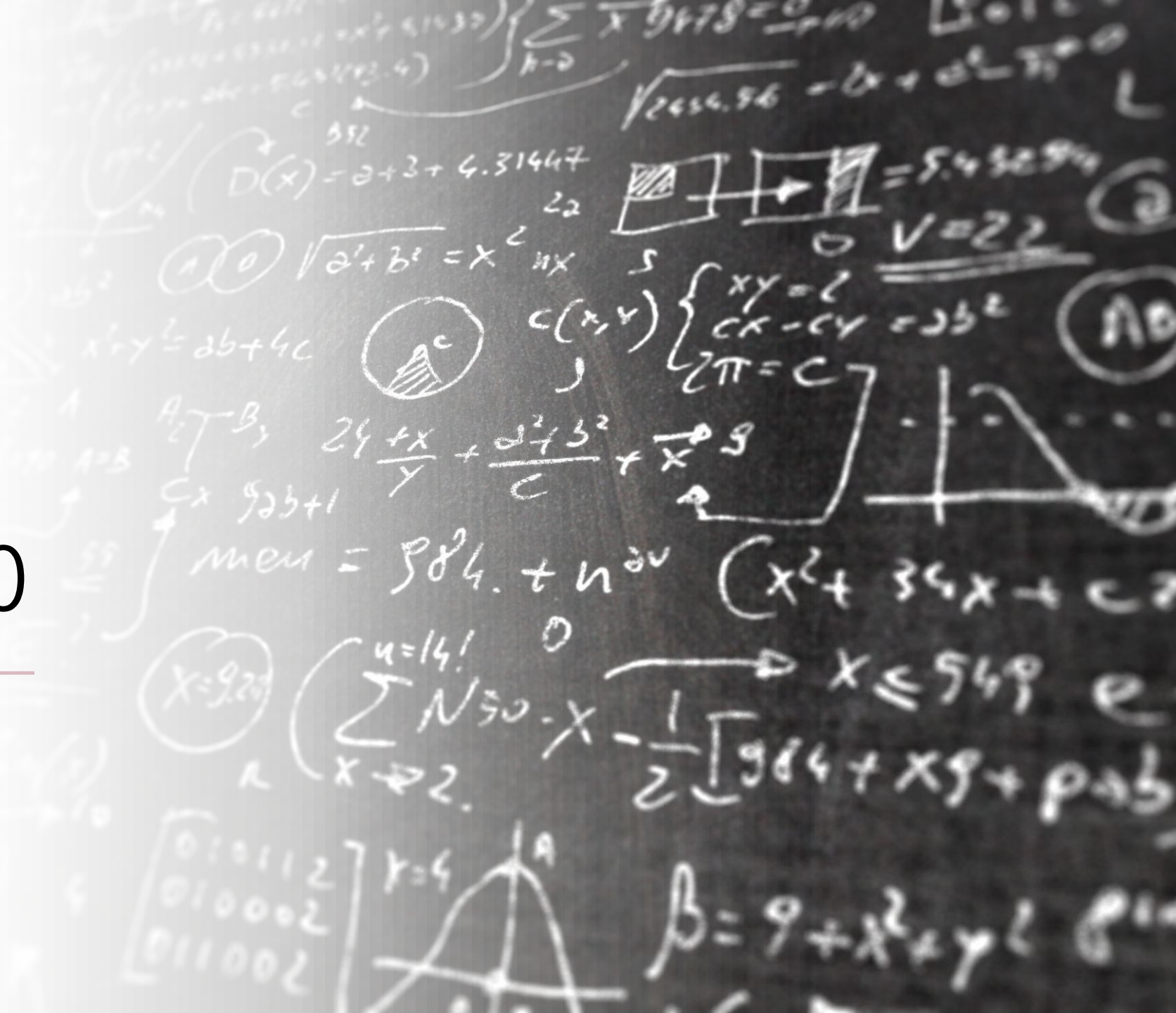
Minimum Loss

- We want to move towards a point with minimum loss!
- It is not the best fit, but it is better than nothing!

Minimum Loss

- Mathematically speaking, when we want the minimum of something what did we do?

Derivative = 0



Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

Solve Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

$60a + 20b + 56 = 0$

$20a + 8b - 20 = 0$

Solve Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

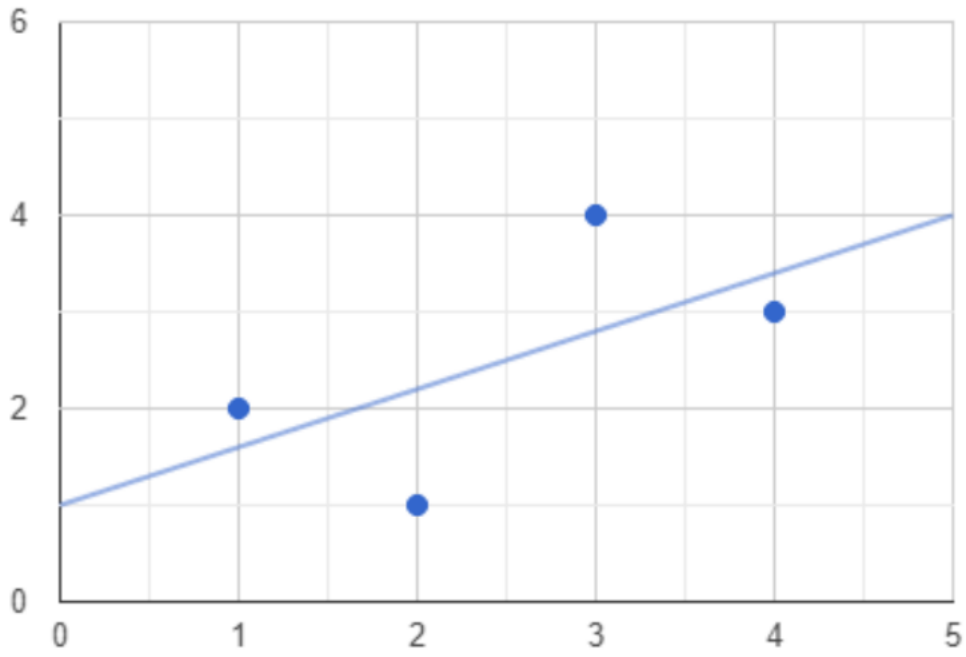
$$\boxed{\begin{array}{l} 60a + 20b + 56 = 0 \\ \\ 20a + 8b + 20 = 0 \end{array}} \Rightarrow \begin{array}{l} a = 1 \\ b = 0.6 \end{array}$$

Final Equation

Now we have a line with minimum error

$$y = x + 0.6$$

Final Output



$y = x + 0.6$

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error


dataset = pd.read_csv("S12_student_scores.csv")

X = dataset.iloc[:, :-1]
y = dataset.iloc[:, 1]

plt.scatter(X, y)
plt.title("Hours vs Marks")
plt.xlabel("Hours")
plt.ylabel("Marks")
plt.show()

X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)

regressor = LinearRegression()
regressor.fit(X_train, y_train)

print(regressor.coef_)
print(regressor.intercept_)

y_pred = regressor.predict(X_test)
df = pd.DataFrame({'Actual':  y_test, "prediction": y_pred})

print(f"MAE: {mean_absolute_error(y_test, y_pred)}")
print(f"MAE: {mean_squared_error(y_test, y_pred)}")
```